

## SRAKAEW, Thailand

WMO: 484400

Lat: 13.7920N

Lon: 102.0325E

Elev: 142

StdP: 14.62

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 62.2       | 65.1 | 54.4                          | 63.4 | 69.3 | 57.3 | 70.3 | 71.7 | 12.6                  | 82.9 | 11.4 | 84.0 | 0.3                      | N/A  | 0.313 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 18.0                   | 99.5            | 80.9 | 97.6 | 80.2 | 95.8 | 79.9 | 83.3                | 93.8 | 82.6 | 92.1 | 81.9 | 90.8 | 2.7                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.9                            | 161.2 | 87.6 | 80.2 | 157.6 | 86.5 | 79.8 | 155.7 | 86.0 | 47.4          | 93.9 | 46.5 | 92.4 | 45.9 | 91.1 | 89.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB    | WB |
| (a)               | (b)  | (c) | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  |       |    |
| (4) 9.8           | 7.5  | 6.2 |                            | 57.6 | 101.2              | 3.1 | 2.5                                                | 55.4 | 103.0      | 53.5 | 104.5      | 51.8 | 105.9      | 49.5 | 107.7 |    |
| (5)               |      |     |                            | 55.6 | 84.5               | 3.3 | 1.8                                                | 53.3 | 85.8       | 51.4 | 86.8       | 49.5 | 87.8       | 47.1 | 89.1  |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 83.1   | 78.8  | 82.2  | 85.3  | 86.8  | 86.3  | 84.6  | 83.1  | 83.2  | 82.9  | 83.0  | 81.8  | 79.0 |
|                                                                                  | DBStd       | 3.78   | 4.08  | 3.51  | 3.34  | 3.10  | 2.65  | 2.25  | 2.18  | 1.89  | 2.06  | 1.97  | 2.67  | 4.03 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 12072  | 893   | 902   | 1094  | 1104  | 1126  | 1038  | 1026  | 1028  | 986   | 1022  | 955   | 898  |
|                                                                                  | CDD65       | 6597   | 428   | 482   | 629   | 654   | 661   | 588   | 561   | 563   | 536   | 557   | 505   | 433  |
|                                                                                  | CDH74       | 71601  | 4239  | 5376  | 7575  | 8022  | 7891  | 6633  | 5880  | 5816  | 5369  | 5671  | 4997  | 4132 |
| CDH80                                                                            | 30910       | 1820   | 2626  | 3905  | 4181  | 3720  | 2718  | 2130  | 2082  | 1842  | 2100  | 2041  | 1745  |      |
| Wind                                                                             | WSAvg       | 1.0    | 1.4   | 1.2   | 1.3   | 1.0   | 0.9   | 1.0   | 0.8   | 0.8   | 0.7   | 0.8   | 1.1   | 1.3  |
| Precipitation                                                                    | PrecAvg     | 57.2   | 0.4   | 1.1   | 1.9   | 4.0   | 6.3   | 7.1   | 8.5   | 8.6   | 11.2  | 7.0   | 1.1   | 0.2  |
|                                                                                  | PrecMax     | 77.8   | 1.7   | 4.1   | 4.8   | 9.4   | 11.1  | 14.6  | 14.2  | 11.6  | 20.7  | 14.5  | 3.5   | 1.2  |
|                                                                                  | PrecMin     | 43.6   | 0.0   | 0.0   | 0.0   | 1.5   | 2.3   | 2.9   | 5.7   | 4.2   | 7.3   | 2.1   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 8.4    | 0.4   | 1.0   | 1.4   | 2.2   | 2.2   | 2.5   | 2.3   | 2.2   | 3.4   | 2.7   | 0.9   | 0.4  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 94.5  | 97.5  | 101.9 | 103.8 | 100.7 | 97.9  | 94.9  | 93.5  | 93.6  | 93.6  | 94.2  | 94.7 |
|                                                                                  |             | MCWB   | 76.3  | 78.1  | 80.7  | 83.8  | 82.4  | 81.4  | 80.7  | 81.5  | 82.1  | 80.5  | 79.2  | 78.1 |
|                                                                                  | 2%          | DB     | 92.7  | 95.6  | 98.9  | 99.9  | 98.0  | 94.8  | 92.5  | 91.8  | 91.8  | 92.1  | 92.8  | 92.7 |
|                                                                                  |             | MCWB   | 76.2  | 77.4  | 79.2  | 81.8  | 81.1  | 81.4  | 80.9  | 81.1  | 81.4  | 80.2  | 78.8  | 77.4 |
|                                                                                  | 5%          | DB     | 90.9  | 94.0  | 96.8  | 97.6  | 95.9  | 92.5  | 90.7  | 90.4  | 90.3  | 90.8  | 91.3  | 90.9 |
|                                                                                  |             | MCWB   | 75.2  | 77.0  | 78.7  | 81.2  | 81.3  | 81.2  | 80.6  | 80.8  | 80.9  | 79.8  | 78.3  | 76.6 |
|                                                                                  | 10%         | DB     | 89.1  | 92.0  | 94.6  | 95.3  | 93.5  | 90.7  | 88.9  | 88.8  | 88.5  | 89.3  | 89.6  | 88.9 |
|                                                                                  |             | MCWB   | 74.5  | 76.6  | 78.8  | 80.6  | 81.2  | 80.9  | 80.3  | 80.5  | 80.5  | 79.6  | 77.9  | 75.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 80.2  | 80.5  | 82.6  | 85.4  | 84.2  | 83.2  | 82.6  | 82.8  | 82.9  | 82.5  | 81.7  | 80.4 |
|                                                                                  |             | MCDB   | 89.9  | 92.7  | 98.0  | 100.7 | 95.3  | 91.0  | 90.0  | 90.0  | 91.0  | 90.1  | 89.9  | 91.1 |
|                                                                                  | 2%          | WB     | 79.0  | 79.6  | 80.9  | 83.3  | 83.0  | 82.5  | 81.7  | 81.9  | 82.0  | 81.6  | 80.8  | 78.9 |
|                                                                                  |             | MCDB   | 88.2  | 91.3  | 94.1  | 96.8  | 93.4  | 90.9  | 89.0  | 89.0  | 89.4  | 89.1  | 89.1  | 88.7 |
|                                                                                  | 5%          | WB     | 77.8  | 78.9  | 80.1  | 82.0  | 82.2  | 81.9  | 81.1  | 81.4  | 81.4  | 80.9  | 80.0  | 77.9 |
|                                                                                  |             | MCDB   | 86.3  | 90.2  | 92.5  | 94.1  | 92.2  | 89.8  | 88.2  | 88.2  | 88.4  | 88.4  | 87.8  | 87.3 |
|                                                                                  | 10%         | WB     | 76.5  | 78.1  | 79.6  | 81.2  | 81.6  | 81.3  | 80.6  | 80.8  | 80.8  | 80.2  | 79.1  | 77.0 |
|                                                                                  |             | MCDB   | 84.5  | 88.3  | 91.4  | 92.3  | 90.5  | 88.7  | 87.6  | 87.4  | 87.3  | 87.2  | 86.3  | 85.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 20.4  | 20.0  | 18.9  | 18.0  | 15.2  | 13.1  | 12.0  | 12.2  | 12.0  | 13.0  | 15.6  | 19.0 |
|                                                                                  |             | MCDBR  | 21.7  | 21.7  | 21.8  | 20.4  | 18.1  | 15.5  | 14.4  | 14.0  | 14.3  | 15.1  | 17.6  | 20.1 |
|                                                                                  |             | MCWBR  | 7.0   | 6.2   | 5.3   | 5.7   | 4.7   | 4.6   | 4.7   | 4.9   | 5.2   | 4.8   | 5.4   | 6.6  |
|                                                                                  | 5% WB       | MCDBR  | 17.9  | 19.2  | 19.5  | 19.3  | 16.0  | 13.9  | 13.1  | 13.1  | 13.5  | 13.9  | 15.2  | 18.0 |
|                                                                                  | MCWBR       | 6.0    | 5.6   | 5.4   | 6.1   | 5.0   | 5.0   | 5.2   | 5.2   | 5.5   | 5.1   | 5.3   | 6.2   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.537  | 0.624 | 0.645 | 0.582 | 0.498 | 0.478 | 0.472 | 0.474 | 0.494 | 0.538 | 0.487 | 0.477 |      |
|                                                                                  | taud        | 1.951  | 1.766 | 1.758 | 1.945 | 2.223 | 2.276 | 2.294 | 2.272 | 2.206 | 2.051 | 2.181 | 2.158 |      |
|                                                                                  | Ebn at Noon | 242    | 226   | 226   | 240   | 257   | 259   | 261   | 264   | 260   | 246   | 255   | 256   |      |
|                                                                                  | Edh at Noon | 58     | 72    | 74    | 62    | 46    | 43    | 42    | 44    | 47    | 54    | 45    | 46    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1569   | 1632  | 1663  | 1763  | 1656  | 1579  | 1506  | 1496  | 1449  | 1488  | 1539  | 1534  |      |
|                                                                                  | RadStd      | 98     | 75    | 82    | 84    | 91    | 66    | 84    | 90    | 73    | 93    | 95    | 62    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (27 neighbors) | N/A   | N/A     | N/A    | N/A     | +0.36 | +0.36 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page